

JetBrains Report 2024

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

Key insights from the annual survey of 23,262 developers worldwide, offering a comprehensive snapshot of programming language usage, tooling preferences, AI sentiment, and employment trends.

Python Closes the Gap: 8 Years of Language Usage Shifts

The top-tier landscape is reshuffling. JavaScript remains #1 but declines from its peak, while Python and TypeScript see massive growth.

Language	2017 Usage	2020 Usage	2024 Usage	8-Year Trend
JavaScript	65%	70%	61%	Declining from peak
Python	32%	55%	57%	Steady growth
HTML/CSS	60%	61%	51%	Gradual decline
Java	47%	54%	46%	Slow decline
TypeScript	12%	28%	37%	Strong growth (Tripled)
Go	8%	19%	18%	Plateaued after growth
Rust	--	7%	11%	Steady growth
PHP	30%	27%	17%	Significant decline

Python Is Now the #1 Primary Language at 35%



The Comfort Zone Effect

Developers show strong loyalty to their chosen ecosystems. Adoption intent for new primary languages remains modestly low across the board.

- Only **6%** plan to adopt Python
- Only **5%** plan to adopt TypeScript
- Only **4%** plan to adopt Java or SQL

Insight: Ecosystem lock-in and expertise investment keep developers loyal. Modest adoption intent signifies few shift their primary stack.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

The strong showing by TypeScript and Rust reflects developer enthusiasm for languages that combine safety, modern tooling, and expressive type systems.



Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Different languages dominate distinct application domains. Language choice is driven by domain, not general preference.

Python

Data & AI

- 31% — AI / Machine Learning
- 28% — Data Science / Analytics
- 38% — Integrating with APIs

TypeScript & JS

User Interface

- 62% — UI Development (TypeScript)
- 58% — UI Development (JavaScript)

Java

App Logic & APIs

- 55% — Application Logic & Workflows
- 49% — APIs & Services

C++

Systems & Hardware

- 27% — System Tools & Components
- 21% — Hardware Interfacing

What This Means for Your Technology Strategy in 2025

1. Invest in Python for Data & AI Teams

Python is the undisputed leader in AI/ML (31%) and Data Science (28%). It's becoming the most critical secondary stack for engineering organizations.

2. TypeScript is the Future of Frontend & Full-stack

TypeScript has tripled its share to 37% and commands the highest developer satisfaction (199). It is the definitive choice for modern web initiatives.

3. Rust and Go are the Growth Plays for Infrastructure

Both are highly prized for performance and safety. Go has stabilized at 18%, while Rust continues its ascent to 11%. Essential for modern systems programming.

4. Re-evaluate Legacy Stacks & Retention Strategy

PHP and Ruby hiring pools are shrinking rapidly. Since top developer satisfaction correlates directly with type safety and modern tooling, modernizing your stack is a core retention strategy.